Confidence Estimation Techniques for white blood cell classification in Healthcare

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***Abstract* - Acute Lymphoblastic Leukemia (ALL) is a serious blood cancer that causes too many immature lymphocytes to form in the bone marrow. Detecting its stages early—beginner, early, pre-leukemic, and pro-leukemic—is important for timely treatment and better outcomes. Traditional methods, which involve manually examining blood smear images under a microscope, can be slow and may lead to different results depending on who is looking. In this project, we propose a deep learning-based framework for automated white blood cell (WBC) classification and confidence estimation to identify the stages of ALL from microscopic images. The methodology involves preprocessing a publicly available dataset of labeled microscopic WBC images, followed by the implementation of (CNNs) for feature extraction and stage classification. To ensure reliability in clinical decision-making, our approach incorporates confidence estimation techniques. This enables the system not only to classify the stages but also to provide confidence levels for each prediction, aiding in trust and interpretability. Performance metrics will be evaluated to validate the model. This study aims to assist medical professionals with a robust, interpretable, and reliable tool for ALL diagnosis, ultimately contributing to faster and more accurate clinical workflows in hematopathology. Future scope includes expanding the model to multi-class cancer detection and real-time clinical deployment.**

***Keywords*— *Tensorflow, Python, Confidence estimation, softmax probability, Image Classification*.**

I. INTRODUCTION

B-acute lymphoblastic leukemia (B-ALL) is a frequent type of leukemia with an age-independent incidence in people but is still more often diagnosed in children [7,8]. Timely diagnosis of malignant lymphoblasts provides an essential opportunity for the best possible treatment and prognosis, so timely identification is crucial. Invasive diagnostic analysis of the bone marrow with bone marrow aspiration (BMA) and additional analysis with bone marrow biopsy (BMB) are recognized as the gold standard in the diagnosis of leukemia, followed by immunophenotyping and analysis with flow cytometry. Although these techniques provide reliable diagnostic analysis at a high standard of care there are many difficulties associated with process including discomfort for the patient, the amount of time taken from a professional, and finances that require significant costs of healthcare resources. The invasive natureof performing procedures also adds further complications to diagnosis for each of these patients (physical discomfort for the individual and psycho-social distress) to consider, especially given that most patients affected by B-ALL are children.

The microscopic examination of peripheral blood smears as a primary, non-invasive, diagnostic tool has significant drawbacks. Peripheral blood smears are comprised of many cells that must be examined individually, making the process labor intensive and susceptible to human error . Fatigue and the variability of expertise of laboratory personnel and morphology of malignant lymphoblasts compared to benign hematogones increases the chances of misclassification. Hematogones are normal B cell precursors and very similar morphologically to malignant B-ALL cells, but they are benign and require no treatment. The fact that they are indistinguishable with only morphology generally leads to confirmatory testing via flow cytometry in order to differentiate between the two cell types. This step does not only lengthen the time of diagnosis, it increases healthcare costs and dependence on resources.

In recent years, advancements in (AI) and, more recently, in deep learning (DL) methods, have produced promising candidates for analyzing medical images. By utilizing convolutional neural networks (CNNs), a certain class of DL algorithms, we have found promising applications in hematological malignancies. These DL algorithms have automated the task of learning and discriminating salient features from features of an image without the need for handcrafted descriptors [10,11]. In hematological malignancies, CNNs have identified leukemia blasts with accuracy that exceeds those of trained professionals. However, still, the majority of studies remain either experimental or proof-of-concept, and little evidence can be found of these studies translating into practical clinical or point-of-care applications.

In order to overcome these obstacles, this study presents a lightweight CNN-based model for the classification of blood smear images into benign (hematogones) and malignant classes, in addition to identifying B-ALL subtypes (Early Pre-B, Pre-B, and Pro-B). One main contribution of this study is the deployment of the trained model This providing a quick and reliable classification in real-time, thus reducing its reliance on invasive diagnostic options and associated delays with respect to diagnosis.

TABLE 1 B-ALL'S CURRENT DATASET AND ITS SUBDIVISIONS

Types	Subtypes	Sample	Image size
Benign	Hematogones	512	1024 ×768
Malignant	Early Pre-B ALL	979	1024 ×768
	Pre-B ALL	955	1024 ×768
	Pro-B ALL	796	1024 ×768
		3242	

The proposed method is further validated by a locally developed dataset consisting of 3,242 Pictures of peripheral blood smears gathered under controlled laboratory conditions. Each of the images were produced from standard staining procedures and taken using equivalent equipment, thereby ensuring consistency and reliability across the dataset. There were preprocessing steps that were applied to the data including image resizing and highlighted the clinically relevant regions through segmentation and data augmentation in order to endow the model with increased robustness. In specifically, the segmentation of lymphoblasts is used to divide lymphoblasts from the background population elements (erythrocytes and plasma) could ensure improved classification accuracy.

II. LITERATURE SURVEY

Table I presents a summary of key papers relevant to title, including their methodologies and the authors' contributions

Saidani et al. (2024) present a new deep learning framework that employs an optimized CNN and employs extensive image pre-processing and a robust image augmentation strategy to classify five types of white blood cells. Their model performed extremely well (99.86% accuracy) and was efficient with minimal computation, surpassing the performance of several transfer learning models [1].

Katar et al. (2023) offered a Vision Transformer model for white blood cell classification and localization that was also explainable. They used Score-CAM to aid in explainability. Their model achieved over 99% accuracy on both multi-class and binary tasks while providing clinical interpretable information and was robust against imbalanced datasets [2].

An emerging hybrid method that applies both Vision Transformer's and CNN's not only achieved both global and local features of WBC imaging at the same time but also was used to enhance accuracy and object interpretability using attention maps, compared to models that just used one method (2025). The hybrid model used BCCD datasets [3].

Islam et al. (2024) deploy an optimized CNN with image pre-processing, and interpretability techniques (e.g., SHAP, LIME and Grad-CAM) for blood cell classification. Their model reaches an accuracy score of 99.12% which facilitates transparency and could support clinical use across a wide variety of platforms [4].

A hybrid CNN-Transformer architecture is introduced (2025) that combines pre-trained CNN layers with Vision Transformer layers, enhanced by attention-based pruning. This strategy demonstrates competitive performance (macro F1=0.98) across each of the blood cell classes and has improved generalizability [5].

Jeneessha et al. (2025) study the infusion of domain knowledge into pre-trained deep learning models, applying some morphological and clinical prior knowledge to white blood cell classification. Their study has demonstrated solid performance on real clinical data and significantly bested traditional transfer learning models [6].

Table 2: Literature Survey

Methodology	Research Gap
Saidani et al., 2024 [] Custom CNN with image resizing, augmentation, and color normalization; compared with transfer learning	Existing studies focus on general WBC classification but lack subtype identification, confidence estimation, and segmentation techniques needed for accurate and explainable leukemia detection in real-world conditions.
O. Katar et al., 2023. Vision Transformer (ViT) with Score-CAM explainability on 5-class and binary WBC datasets	he study achieves good WBC classification using ViT but lacks leukemia-specific segmentation, subtype identification, and confidence estimation for clinical reliability..
Combines ViT and CNN for feature extraction; tested on BCCD dataset with multi-class tasks.	Existing CNN-ViT models for WBC classification lack real-world validation and struggle with data imbalance, reducing generalization.
O. Islam et al., 2024. CNN with image pre-processing; uses SHAP, LIME, Grad-CAM for interpretability; trained on a large dataset.	Existing CNN models lack explainability across diverse datasets, limiting trust and transparency in automated blood cell classification.
Integrates various CNNs and Vision Transformer layers; uses data augmentation and layer pruning.	Hybrid CNN-Transformer models still face challenges with small datasets and high computational cost, affecting practical clinical deploymen

III. METHODOLOGY

The main goal of this work was to build a completely automated method that could be used on pictures for the diagnosis of ALL and the classification of its subtypes (early pre-B, pro-B, and pre-B ALL). Here, a model utilizing DL algorithms is put forth for the classification and categorization of patient samples suspected of having ALL.

The proposed model was trained and its effectiveness evaluated on a local dataset. These photos were taken at various hospitals in Tehran, Iran. Qualified laboratory staff

processed and stained samples of blood from 89 patients accused of having B-ALL, producing 3242 blood microscopic images in total. The dataset contains both benign and malignant classes. Hematogones that closely resemble B-ALL cases are found in the first class (benign), but these hematopoietic precursors are benign, have no need for chemotherapy, and typically go away into the bloodstream on their own. The second class of malignant lymphoblasts (ALL) includes three subgroups. Using flow cytometry, this cell class and its subclasses were identified. Table 1 presents comprehensive details about this dataset after the data has been shuffled.

A. Dataset

In the current investigation, all of the gathered slides were initially stained in a controlled setting by a number of laboratory experts to preserve picture dependability and avoid structural variations brought on by various staining and photography techniques. A microscope, the same Zeiss camera, and a designated hematology specialist were also utilized to take each slide image. To maintain the principles of reliability and consistency, certain protocols have been complied to. In computer-based visual recognition and related applications, the performance models are directly impacted by pre- processing. Pre-processing methods can enhance diagnosis, expedite DL model training, prepare input data, eliminate noise, and emphasize the image's valuable characteristics .

B. Data Preparation

Research data images in PNG or JPG form were decoded, or transformed into numeric arrays based on their pixel values. CNNs frequently reduce the size of input images to maintain network design flexibility and reduce computational load. The inputs in the pre-trained network used for this investigation need to be a specific size. Furthermore, the size of the input pictures should be 224 × 224 in many standard designs, and these dimensions are also used in current medical diagnosis investigations. As thus, 224 × 224 was the new image size.

C. Data segmentation

At this point, the model's key components are shown to divide the ROI regions. To extract and show areas that are clinically meaningful to the medical specialist and, consequently, the ML model, segmentation was used in this investigation. In this work, one of the objectives of segmentation was to remove and disregard extraneous components that are irrelevant for learning models and interfere with learning as noise. In order to improve the accuracy and dependability of the model and sensitive the pathway to lymphoblast cells, the input pictures were segmented using a special technique that distinguished the cells of the blast from other blood components (erythrocytes and plasma).

We employed an effective and economical method for segmenting blast cells that depends on color period division and color thresholding. Blood microscopic pictures in the RGB space by default have three red, green, and blue channels. Here, image segmentation using the grouping paradigm was used. A strong clustering algorithm in the L*a*b color space of pixels is used in this segmentation technique. By aligning every pixel with the appropriate cluster, this technique immediately segments images. The International Commission on Illumination provided the specifications for this color space, which CAE first established. Based on nonlinearly compressed CIE XYZ color space coordinates, a lab color space is a color- opponent structure with dimension L for lightness and a and b for the color opposing dimensions. The Hunter 1948 L, a, b color space has the following coordinates: L, a, and b. The goal of the non linear relations for L*, a*, and b* is to replicate the eye's nonlinear response. Additionally, consistent component changes in the L*a*b* color space are intended to match consistent changes in perceived color. Therefore, by considering each color as a point in a three- dimensional (3D) space (which has three components, L*, a*, and b*), and calculating the Euclidean distance between them, one may estimate the relative perceptual differences between any two colors in L*a*b*[8,9] The procedure of data segmentation is shown in fig 1 and fig 2.To convert the pictures from the RGB space, represented by (XYZ), into the LAB space, transformations 1 through 3 are needed:

$$L^* = 116f\left(\frac{Y}{Y_n}\right) - 16 \tag{1}$$

$$a^* = 500\left[f\left(\frac{X}{X_n}\right) - f\left(\frac{Y}{Y_n}\right)\right] \tag{2}$$

$$b^* = 200\left[f\left(\frac{Y}{Y_n}\right) - f\left(\frac{Z}{Z_n}\right)\right] \tag{3}$$

after converting RGB to LAB Color space, Because of its emphasis on blast color and increased ability to distinguish between the blast and backdrop, the A component was taken out of the LAB space, and the original image output was B. The image results were then obtained as C through the use of the K-means clustering. Binary thresholding was then applied to the image result (Fig.1.Image d), blurring or ignoring a large number of superfluous background elements. At this point, the image output had diagnostically insignificant components; therefore, a mask had to be placed to the images in order to clean and eliminate their backgrounds and extract the blast features while ignoring other features. At this point, the image output had diagnostically insignificant components; therefore, a mask had to be placed to the snap in order to remove and eliminate their backgrounds and extract the blast features while ignoring other features. Consequently, no extraneous component of blood or cell was present in the picture outputs (Fig.1.picture e) following the application of the mask. Ultimately, ROI areas were retrieved by segmenting the original images and mapping this result to the initial images

The blast cells were isolated from the input photos by applying this mask.

D. Data rescaling, Augmentation and Categorization

The applicable values of the snaps needed to be shown in a single time frame in order to display the images in the same form. The input data was homogenized at this stage using normalization of pixel intensity in the [0,1] range. It's insolvable to overdo how important data addition is to the performance of the ML- grounded model. It comprises altering images to save natural and semantic information while adding the friction of training samples.

Table 3 Splitting of dataset into training, validation and testing set

	Training Dataset	Validation Dataset	Testing Dataset	
Sample	5250	584	325	6159
Percentage (%)	85	10	5	100

Two transformations are used in training samples to alter the filmland that depict the variation of real- world data. These changes correspond of both vertical and perpendicular flips shown in (table 3). The data demanded to be distributed in order to train both the suggested model and the pre-trained networks.

E. Convolution Neural Network structure

One of the key characteristics of effective models was their reduced weight, since the objective was to employ a good-turning model in less-specific tools. In order to choose one of these pre- trained model as the extraction of features block, we used Four lightweight algorithms and evaluated their effectiveness and light weight. The networks chosen for this investigation wereVGG-19, ResNet50V2, MobileNetV2, and EfficientNetB0 as classification model as shown in fig 5.

F. EfficientNetB0

The EfficientNetB0 architecture served as the basis for extraction of features in our work, and we began to build the model with pre-trained ImageNet weights. By selecting

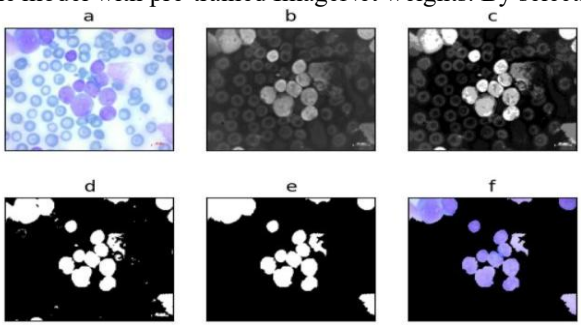

Fig. 1. Data Segmentation stages: a) RGB color space, b) RGB to LAB color space, c) K-mean Clustering, d) Binary Thresholding, e)Mask Generation f) Applied mask on the RGB image

include_top=False, which removed the fully linked classification layers, we were able to optimize the model specifically for the job of identifying (ALL) from (WBC) pictures. The input photos were downsized to 224×224×3 to correspond with the expected network dimensions in order to guarantee consistency across the dataset.

EfficientNetB0 was selected due to its compound scaling technique, which achieves a robust performance at a relatively cheap computational cost by balancing network length, breadth, and dimension. Using ImageNet's transfer learning improved generalization and accelerated training. because the model could reuse learned low-level and mid-level characteristics while modifying its more complex representations for the medical field.

G. MobileNetV2

MobileNetV2 was also taken into consideration as a potential backbone in this case study because of its lightweight design and effectiveness in managing picture categorization jobs. The model may exploit previously learnt visual features and adapt them to the domain of (WBC) analysis by using pre- trained ImageNet weights and scaling the input images to 224×224×3. Because of its small size, MobileNetV2 is especially well-suited for applications in medicine where processing power may be scarce.

H. VGG-19 and ResNet50V2

The efficacy of ResNet50V2 and VGG19 in categorizing white blood cells for the identification of ALL was assessed. The problem with vanishing gradients is lessened and deeper networks can be trained thanks to residual connections in ResNet50V2, which has about 7.2 million parameters. It is ideal for identifying fine-grained cellular patterns because of its skip connections, which improve feature propagation. one of the first and most popular CNNs for medical imaging tasks is VGG19, which uses a simple design of 19 convolutional and fully linked layers. VGG19 provides a solid baseline for performance comparison since it effectively captures hierarchical picture features despite its size and processing requirement

I. Selective structure, train and evaluation

After assessing various fitting methods and designs, we decided to use segmented pictures to better target lymphoblast cells and improve accuracy and dependability. To choose the suggested model, pre-trained models were assessed after pre-processing, segmenting the dataset. Pre-trained models did not achieve acceptable performance metrics After consulting with hematologists, we determined that the weak results could be attribute to incomplete picture feature extraction. The model was developed to use both segmented and original photos in its networks and give various accuracy that shown in Table 4.

To maximize feature space, selected models were fed both the original and segmented image channels. Collecting images from two channels allows for a comprehensive coverage of study image features.

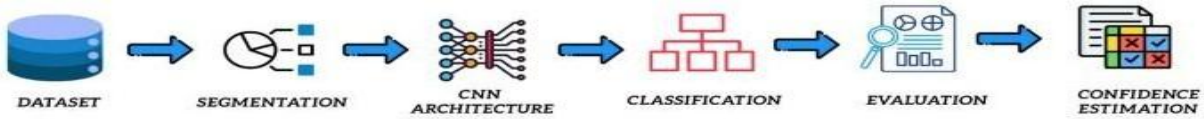

Fig.2. Overview of Basic model Structure

Table 4. Performance of the chosen network in a specific context

Algorithm	Epoch	Training Accuracy	Batch size	Size (MB)	Parameter	Val accuracy
EfficientNetB0	30	99.23	32	18.525	2,428,100	99
MobileNetV2	30	99.51	32	17.30	2,268,228	98
VGG19	15	98	32	17.30	2,268,228	89
ResNet50V2	15	99	32	55.01	7,210,445	99

J. Metric evaluation

This study evaluated the proposed identification and categorization model using medically pertinent statistical parameters. Suggested measurements include specificity, recall, and accuracy.

The following parameters were required to calculate these evaluation metrics: True Positive (TP): the number of correct predictions of positive cases by the method; True Negative (TN): the number of correct predictions of negative cases; ; False Positive (FP): the number of incorrect predictions of positive cases; and False Negative (FN): the number of incorrect predictions of negative cases by the method However, in the situation of an imbalanced dataset, accuracy is not guaranteed. Accuracy is the optimal statistic for evaluating model performance. However, in this study, accuracy metrics were used to choose the most efficient model based on pre-trained networks.

Accuracy: Accuracy measure a method's competence by calculating the proportion of correctly identified cases compared to the total amount of cases. It may be stated mathematically as [11]:

$$Accuracy = \frac{TN+TP}{TN+TP+FN+FP} \quad (4)$$

Recall: The proportion of accurately anticipated positive observations to all of the actual class observations [11].

$$\text{Recall} = \frac{TP}{TP+FN} \quad (5)$$

F1 Score: The F1 score is an effective metric for classification challenges on imbalanced sets of data since it is more vulnerable to data distribution.

$$\text{F1 Score} = 2 * \frac{Precision*Recall}{Precision+Recall} \quad (6)$$

IV. Result

In this part, we initially evaluate the accuracy of the pretrained neural networks on the training and test datasets, selecting the most effective model as the chosen model. Following hyperparameter adjustment, the network's efficiency is measured using the test dataset.

A. Efficient model selection

After training the models on the training set and evaluating them on the test set for up to 30 epochs, the performance of the pre-trained networks was compared under identical conditions. The evaluation metrics for each model are summarized in Table 4, allowing for a systematic comparison of their efficiency. This investigation revealed that while all models displayed adequate accuracy, MobileNetV2 offered the optimum balance of performance and efficiency. It obtained great classification accuracy with much fewer parameters, smaller model size, and lower layer implementation on systems with limited hardware resources. As a result, MobileNetV2 was determined as the most effective model for this study and chosen for the final deployment.

B. Performance of the selected model

Fig. 3 shows the learning curve of the chosen model during the training phase with the hyperparameters for the training and validation sets adjusted. The suggested model's performance with fresh validation data showed that, as the number of epochs increased, the model's error decreased and its accuracy on unknown data improved. This suggests that the model has a good capacity for generalization in identifying novel instances of blood microscopic images. The suggested model was able to get maximal coverage of discriminative features by using both original and segmented photos as input, guaranteeing that every sample in the test dataset was correctly identified. Following thorough testing, the selected pre-trained network (such MobileNetV2 or EfficientNetB0) showed exceptional accuracy and efficiency, and as a result, it was chosen as the study's final backbone design.

C. Confidence Estimation

Confidence estimation methods were used to further improve the suggested categorization model's dependability. One of the post- processing calibration techniques used to modify the network's estimated probabilities was temperature scaling. This method smoothes the probability distribution by lowering over-confident predictions and producing more calibrated confidence scores by adding a temperature parameter ($T > 1$) to the softmax function. This guarantees that the actual likelihood of correctness is more accurately reflected in the anticipated probability values.

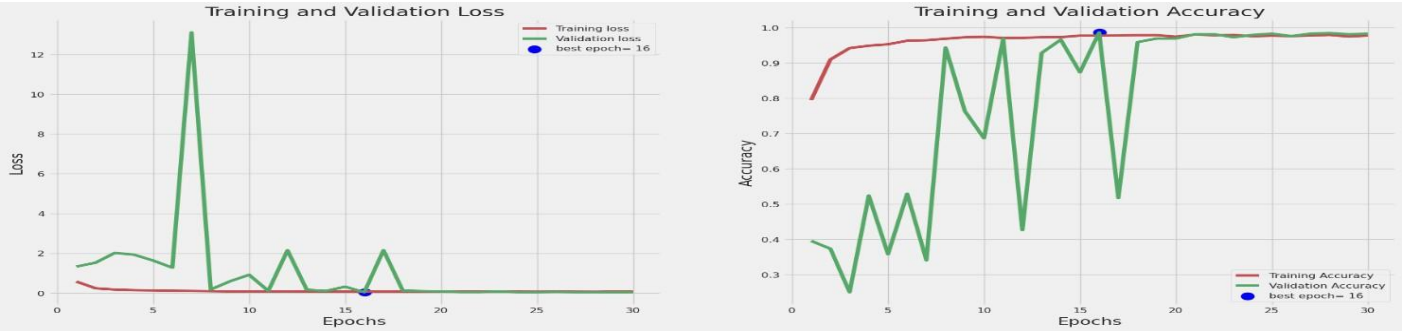

Fig 3. The proposed model's training and validation accuracy, loss during training. The model reached convergence with training accuracy and loss of 100% and 0.0035, respectively, after 16 epochs.

Temperature scaling on the validation set improved the model's interpretability and reliability for clinical use by producing accurate classifications together with a calibrated prediction confidence measure.

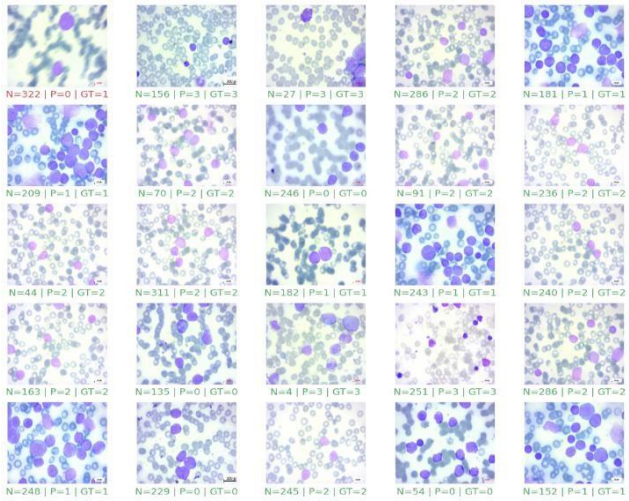

Fig.4. The model's performance was assessed using nine randomly selected blood microscopic samples. N, P, and GT stand for image index, model prediction, and groundtruth label, respectively. A correct forecast is indicated by a green caption, whereas a wrong prediction is shown by a red caption.

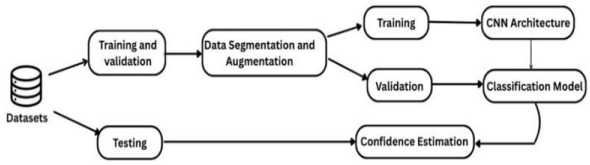

Fig.5. Block Diagram

V. Conclusion

This study describes a comprehensive and sophisticated diagnostic system for automatically classifying (ALL) phases using (WBC) pictures. The method is intended to precisely detect the minute morphological shifts that take place throughout the four crucial stages of ALL—beginning, early, pre, and pro—by utilizing the abilities of deep learning, especially transfer learning with CNN architectures. To increase prediction reliability, the model uses sophisticated training tactics including data augmentation, optimization approaches like SGD with momentum, and confidence calibration utilizing techniques like temperature scaling. This improves the model's generalizability across various imaging conditions and patient demographics in addition to guaranteeing excellent accuracy and sensitivity in early-stage identification. Additionally, the suggested full-stack architecture combines an intuitive web interface with a strong backend driven by Gradio, Flask/FastAPI, Streamlit and Tensorflow, allowing for real-time communication and accessibility for physicians.

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